



INTERNAL COACH EDUCATION • VERSION 1.0

THE FOB COACH SCIENCE COMPENDIUM

A chronological explanation of what the system is, what the body is managing, how the scoring works, and what we can honestly say so far.

Not marketing. Not a diagnosis. Not finished science.

A working education resource built from prototype testing, current session rules, FOBbreakdown logic, and the evolving FOB research framework.

READ THIS IN ORDER

Each concept builds on the one before it. Start with the equipment, then the motion, then the body, then the scoring. Do not jump straight to a score interpretation without checking the conditions that produced it.

ESTABLISHED PRACTICE

A procedure we can standardize now: equipment inspection, anchor setup, session order, scoring rules and coaching progressions.

OBSERVED PATTERN

Something repeatedly noticed during prototype use: rhythm demand, torso fatigue, grip demand, heart-rate rise or loss of control.

WORKING HYPOTHESIS

A plausible adaptation or transfer effect that still needs formal testing. Say “may,” “suggests,” or “we are testing.”

DO NOT BLUR THEM

A clean coach explains what was directly measured, then separately explains what it might mean.

WHAT FOB ACTUALLY IS

FOB is a user-driven resistance system. A moving ballast is suspended within an elastic band. The user starts the oscillation, receives the returning force, redirects it, and keeps the cycle alive.

WORKING DEFINITION

A suspended ballast oscillates through repeatable directions and creates rapidly changing pulling demands that the user manages through position, rhythm, force and control.

The user is not just reacting to random disturbance. The user is also the energy source. That is what makes the system different from passive instability tools.

THE PRODUCT FORMULA

FOBand

The elastic element. It transmits force, stretches, recoils and changes tension as the ballast moves.

FOBBlock

The organized loaded section. It protects the contact area and carries the selected ballast weight.

FOBallast

FOBand + FOBBlock working as one assembly.

ANCHOR / GRIP

The anchor fixes the origin of pull. The grip is where the user receives and applies force.

COACH RULE

Never discuss “the weight” by itself. Band level, block weight, distance, threshold and duration all change the task.

WHAT CREATES THE RESISTANCE?

The user feels a combined demand, not one single force.

ELASTIC PRELOAD

The band begins under tension so the system is active instead of slack.

ELASTIC RECOIL

The band stretches and returns as the ballast travels.

INERTIA

The ballast resists a change in motion.

MOMENTUM

Once moving, the ballast must be redirected rather than simply stopped.

ACCELERATION

Faster changes in speed or direction require more force.

GEOMETRY

Anchor height, user distance, movement angle and orientation change the line of pull.

INITIATE. RECEIVE. REDIRECT. REPEAT.

This is the basic FOB cycle. Every clean repetition helps create the next one.

1

INITIATE

Create the first useful motion.

2

RECEIVE

Accept the returning pull without losing position.

3

REDIRECT

Send the force into the intended path.

4

REPEAT

Keep rhythm without adding random movement.

NO PASSIVE PHASE

The demand does not disappear between repetitions. The system is still pulling, returning or changing direction.

WHO IS CONTROLLING WHOM?

The most important coaching question is simple: is the user controlling the oscillation, or is the oscillation controlling the user?

CONTROLLED

The intended path remains clear. Stance stays organized. The user can stop on command.

UNCONTROLLED

The ballast drags the body, the path widens without intention, feet chase the load, or the user cannot stop cleanly.

CORRECTION ORDER

Reduce amplitude, reduce threshold, lower resistance or weight, shorten duration, simplify stance.

PROGRESSION ORDER

Build duration first. Then add amplitude, resistance, stance variation, orientation or speed.

WHY RHYTHM MATTERS

Rhythm is not decoration. It is how the user repeatedly times force so the returning load becomes useful instead of disruptive.

FOB ECONOMY

A coaching concept for using less unnecessary effort to maintain the same useful oscillation. It is not yet a validated physiological measurement.

A smoother user may look less dramatic while producing a better session. The coach should reward repeatable timing, not theatrical movement.

WHAT THE BODY IS MANAGING

TRUNK ORGANIZATION

The torso keeps the rib cage, pelvis and spine organized while force changes direction.

SHOULDER + ARM ACTION

The upper body produces, receives and redirects force through the chosen path.

STANCE

The lower body supplies a stable base and may contribute controlled rotation at higher levels.

BREATHING

Breathing should continue. Breath-holding may be brief under demand, but it should not become the main strategy.

GRIP

The hands connect the user to a changing pull. Grip fatigue can limit the task before the core does.

ATTENTION

The user must continually track timing, direction, marker position and body control.

THE CATEGORY WE ARE BUILDING

Reactive Endurance is the ability to keep producing and controlling useful movement while resistance keeps returning and changing direction.

PLAIN LANGUAGE

How long can the person keep moving well before tiredness changes the movement?

It is not just strength. It is not just cardio. It is not just balance. It is repeated force production plus repeated force control under fatigue.

WHAT THE SCORE IS TRYING TO DESCRIBE

DURABILITY

How long good movement lasts during long holds and sustained work.

FATIGABILITY

How quickly quality drops once the user is working.

REPEATABILITY

How well performance returns after a short break and another round.

RESILIENCE

How well the user stays organized when direction, position or demand changes.

IMPORTANT

These are FOB performance constructs. They organize session observations. They are not medical diagnoses or universal laboratory traits.

THE FOUR PUBLIC DIRECTIONS

Direction is part of the test. A score has no meaning without knowing where the force came from and where the user had to send it.

FRONT

Force is organized in the forward-facing setup.

LEFT

The user works with the line of pull biased to the left.

RIGHT

The user works with the line of pull biased to the right.

OPPOSITE / REVERSE

The user faces or acts against the original setup to change the demand.

Current scoring commonly compares front, left and right. Side-to-side gaps matter because an identical total can hide directional weakness.

WHAT EACH SCORED SECTION ASKS

REACTIVE CORE

Twist, brace and control returning force while the torso stays organized.

REACTIVE ISOMETRIC

Hold a chosen position while the moving system actively tries to pull the body away from it.

REACTIVE PUSH / PULL

Send force out and receive it back repeatedly without losing the path.

TECHNICAL MOVEMENTS

Apply the same oscillatory logic to curl/extension, lateral raise and upright row/pressdown patterns.

REACH. HOLD. RECOVER.

REACH Reach today's FOB Marker.

HOLD Maintain the oscillation outside the marker.

RECOVER If the movement comes back inside, restore the marker within the recovery window.

WHY THE MARKER EXISTS

The marker turns “that looked good” into a repeatable boundary. It gives the coach a visible threshold for amplitude and sustained output.

WHAT A POINT ACTUALLY MEANS

A point is not a direct unit of strength or fitness. It is credit for meeting a specific task rule under specific conditions.

BASE POINTS

Credit for reaching and maintaining the required marker or position.

DROPS

Moments when the movement returns inside the marker and requires recovery.

RECOVERY POINTS

Credit for restoring performance quickly after a drop, when the rule allows it.

TECHNIQUE POINTS

Credit for preserving the defined movement pattern, not merely creating large motion.

LOW ENDURANCE + HIGH RECOVERY POWER

A user can lose many points, then regain points by reaching an upper threshold or recovering quickly. That is not the same as simply performing poorly.

WHAT HAPPENED

Quality could not be maintained continuously, but the user could still produce strong short bursts and restore the target after a mistake.

COACH INTERPRETATION

Immediate power or re-acceleration is present. The main training goal is extending that output so fewer recoveries are needed.

Say: “You can get the movement back. Now we need it to last longer before it drops.”

WHY WE REPEAT THE PROTOCOL

Round 1 shows what the user can do fresh. Round 2 shows how much of that performance remains after work and a short recovery.

$$R2 \approx R1$$

Strong repeatability under the same conditions.

$$R2 < R1$$

The first round created fatigue that was not fully recovered before the second.

$$R2 > R1$$

The user may have learned the task, warmed up further, improved rhythm, or benefited from a condition change.

CONDITIONS CHANGED

Do not call this pure repeatability. Explain exactly what was made easier or harder.

A SCORE WITHOUT CONDITIONS IS INCOMPLETE

THRESHOLD DISTANCE

Farther markers generally require more usable amplitude. A closer marker reduces the required travel.

FOBLOCK WEIGHT

More mass increases inertia and momentum. It may slow reversal while increasing the demand to redirect.

FOBAND LEVEL

More band tension changes preload, recoil and the force required to create useful travel.

INTERVAL DURATION

Longer intervals test how long quality is maintained. Shorter intervals favor fresh output.

DIFFICULTY PRESET

A bundled starting condition. It must be unpacked into actual variables for interpretation.

ROUND CHANGES

Any change between rounds must be named before comparing scores.

CONTEXT CHANGES THE RESULT

Two athletes can both score 80%, but one may have used a closer marker, lighter block, lower band level and shorter interval.

COACH STANDARD

**Report the score and the conditions together.
Never praise or criticize the number without
stating what the person was asked to do.**

The clean comparison is the same person, same movement, same conditions, repeated over time.

HOW LONG GOOD MOVEMENT LASTS

Durability combines the long-hold sections, especially Reactive Core and Reactive Isometric.

HIGH

The end of the interval still resembles the beginning. Position and output remain available late.

LOW

Quality changes early or repeatedly. The user may need a shorter interval, smaller threshold or easier loading condition.

GYM

Longer sets, carries and repeated bracing may stay organized deeper into fatigue.

LIFE

Sustained carrying, standing, walking or physical work may be less likely to force an early change in posture or strategy.

Transfer statements remain hypotheses unless directly tested outside FOB.

HOW FAST QUALITY FALLS

Fatigability is described by the timing and number of drops.

FEW LATE DROPS

Quality lasted through most of the work and changed near the end.

MANY EARLY DROPS

The selected condition exceeded the user's current sustainable output.

RECOVERS EACH TIME

The user can restore output but cannot yet hold it continuously.

DOES NOT RECOVER

The user may need a larger regression because the current condition is no longer productive.

WHAT COMES BACK AFTER REST

Repeatability compares the second exposure with the first.

GOOD REPEATABILITY

A short rest restores enough capacity for the person to perform close to the first round under the same conditions.

This may matter in sports, work and gym tasks that happen in repeated bouts. Do not claim transfer from one FOB session alone. Use it as a coaching hypothesis and track it over time.

WHAT HAPPENS WHEN THE TASK CHANGES

Resilience is the ability to preserve control when the direction, side, stance or demand changes.

HIGH TOTAL + EVEN DIRECTIONS

The user stays organized across more than one line of pull.

HIGH TOTAL + LARGE SIDE GAP

Strong overall performance with a specific directional limitation.

LOW TOTAL + EVEN DIRECTIONS

The condition may be globally difficult rather than side-specific.

IMPROVES AFTER PRACTICE

Motor learning may be contributing. Do not automatically call it physical adaptation.

THE CURRENT 45-60 MINUTE MODEL

01 ASSESSMENT & WARM-UP

8-10 minutes

02 TRAINING ROUND 1

03 TRAINING ROUND 2

Repeat the complete protocol

04 FOBREAKDOWN

Review results and next condition

The coach is not only delivering exercises. The coach is establishing a repeatable test environment, then changing one variable at a time.

THE FIVE QUESTIONS

- What is moving, and what force is the user currently managing?
- What direction and body orientation are being trained?
- Is the user controlling the oscillation, or is it controlling the user?
- Which variable should change next, and why?
- What outcome is this variation intended to develop?

COACHING TONE

Clear, calm and direct. Explain the next action in ordinary words. Avoid riddles, vague motivation and unsupported scientific claims.

WHEN TO REGRESS OR STOP

STOP

Sharp pain, dizziness, illness, loss of awareness, equipment failure, uncontrolled anchor movement or inability to stop the oscillation.

REGRESS

Repeated foot chasing, uncontrolled trunk rotation, loss of grip, breath-holding that does not resolve, or a path too large to control.

INSPECT

Band cuts, abrasion, whitening, deformity, loose stitching, damaged block, weak connectors or a sharp/unstable anchor.

ENVIRONMENT

Use clear space, reliable footing and an anchor that is appropriate for repeated multidirectional force.

THE CURRENT RESPONSIVENESS HYPOTHESIS

Founder field testing rated the 70 lb band with a 10 lb load at 6.5/10. The next prototype uses a 50 lb band, an 8 lb FOBlock and a shorter 22-inch working band section.

EXPECTED CHANGE

Lower tension and lower mass may make reversal quicker and reduce the feeling of fighting the apparatus.

SHORTER LENGTH

May keep the loaded section closer to the useful working range and reduce lag.

WHAT WE ARE TESTING

Flow, sustained oscillation, clean reversal, rhythm, controllability and desire to continue the set.

NOT YET A FACT

The improved experience is a prediction. The prototype must be tested under repeatable conditions.

CURRENT DESIGN PRINCIPLE

For FOB, responsiveness may matter more than maximum resistance.

WHAT WE KNOW VS. WHAT WE ARE TESTING

SUPPORTED BY ESTABLISHED MECHANICS

Elastic tension, inertia, momentum, acceleration and geometry all contribute to the task.

SUPPORTED BY CURRENT PRACTICE

The session can be standardized, scored, repeated and progressed through observable rules.

OBSERVED, NOT FORMALLY VALIDATED

Torso fatigue, grip demand, rhythm improvement, heart-rate elevation and different experiences across loading setups.

WORKING HYPOTHESES

Transfer to sport, work capacity, aging, rehabilitation, hypertrophy, metabolic outcomes and long-term adaptation.

WHAT SHOULD BE MEASURED NEXT

- Reliability: does the same user produce similar scores under the same conditions?
- Validity: what established tests do durability, fatigability, repeatability and resilience correlate with?
- Instrumentation: force, displacement, oscillation frequency, heart rate and movement quality.
- Dose-response: how do band level, block weight, distance and interval duration change the task?
- Learning effect: how much score improvement comes from becoming familiar with the movement?
- Transfer: do changes in FOB performance predict changes in gym, sport or daily tasks?

HOW A COACH SHOULD EXPLAIN RESULTS

OBSERVATION

“You crossed inside the marker three times in the final minute.”

INTERPRETATION

“Your output was available early, but it was harder to maintain late.”

CONTEXT

“Round 2 used a lighter band, so the higher score cannot be treated as pure recovery.”

NEXT STEP

“Keep the same marker and try to delay the first drop.”

NEVER SAY

“Your core is weak,” “you are injury-prone,” or “this proves you will perform better in life.” The score does not support those claims.

WHAT A GOOD FOB COACH DOES

A good coach does not memorize the most movements. A good coach can see the system, establish it safely, explain what is happening, recognize lost control and change one variable for a clear reason.

THE JOB

Create a repeatable experience where the user can generate, receive and redirect force while the coach measures what remains available as fatigue and task changes accumulate.

That is the current scientific and coaching center of FOB Systems. Everything else should build from it.

DOCUMENTS USED TO BUILD THIS EDITION

- FOB SESH Coach Rulebook, Rev. 2.1 - current session order, scoring and coach procedures.
- FOB SESH & FOB Breakdown - current client-facing outcome model and scoring interpretation.
- FOB Framework v1.0 - evidence labels, physics, coaching standard, safety and research roadmap.
- FOBible Expanded First Edition - plain-language system definitions and chronological conceptual development.
- JahFob: The Reversal Method v3.0 - earlier apparatus mechanics and historical research framing.
- FOB SITE v114 - current terminology and ecosystem implementation.
- Founder field notes through August 2026 - current prototype V2 hypothesis and testing priorities.

VERSION RULE

When a newer coach rule or product specification conflicts with an older paper, use the newer operational standard and preserve the older claim only as history.